

Deep Reinforcement Learning

Advanced Algorithms

Prof. Dr. Sebastian Peitz

Chair of Safe Autonomous Systems, TU Dortmund

Summer term 2026

Content

Content

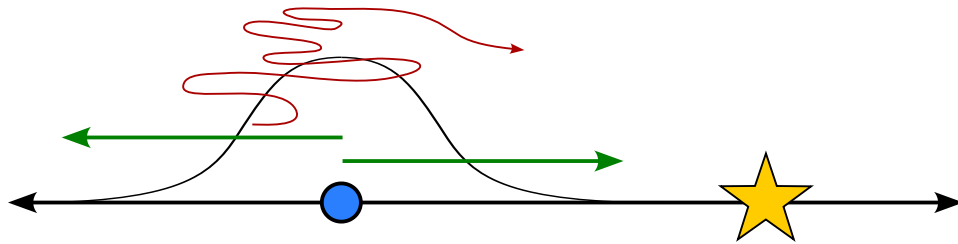
- TRPO / PPO
- DDPG / TD3 / SAC
- Dueling DQN?

Where are we?

Chapter	Topic	Content
	Basics & tabular methods	
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
	Deep-learning-based methods	
6	Brief introduction to deep learning	The basics for what comes next
7	Value function approximation	Value estimation with function approximation
8	Deep Q-learning	Q-learning with neural networks
9	Policy gradients	Direct optimization of the policy
10	Actor-critic algorithms	Improved policy gradients via value functions
11	Advanced algorithms	
	Model-Based Control	
	Advanced Topics	

Natural / covariant policy gradients (TRPO)

Ill-conditioned gradients



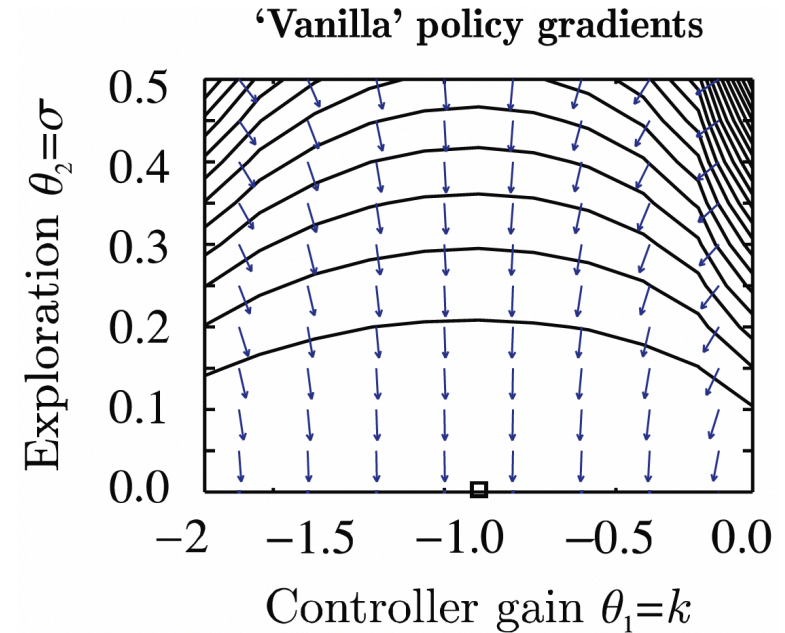
Inspired by Sergey Levine's [CS285 lecture](#).

$$r_t = -s_t^2 - a_t^2$$

$$\log \pi_\phi(a_t|s_t) = -\frac{1}{2\sigma^2}(ks_t - a_t)^2 + C, \quad \phi = (k, \sigma).$$

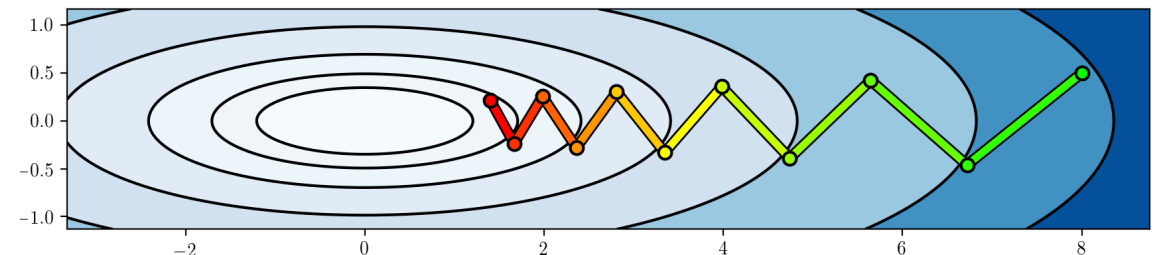
$$\nabla_\phi \log \pi_\phi(a_t|s_t) = \begin{pmatrix} -\frac{(ks_t - a_t)s_t}{\sigma^2} \\ \frac{(ks_t - a_t)^2}{\sigma^3} \end{pmatrix}$$

- **Optimum:** $\phi^* = (-1, 0)$.
- **Issue:** The gradients do not point towards the optimum for smaller σ !
 - The σ^{-3} term blows up.



Source: [\(Peters and Schaal 2008\)](#)

Closely related to ill-conditioned optimization problems:



References

Peters, Jan, and Stefan Schaal. 2008. "Natural Actor-Critic." *Neurocomputing* 71 (7): 1180–90.